

UK Ethnic Demographic Change Under Alternative Migration and Demographic Scenarios, 2022–2122

Phase 1: Methodological Design and System Architecture

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Abstract

This paper presents the design of a transparent, reproducible population-projection system for examining how the self-identified ethnic composition of the United Kingdom could evolve over a 100-year horizon under alternative demographic assumptions. The system implements a multistate cohort-component model with separate treatment of immigration, emigration, fertility convergence (planned), partnership-weighted fertility inputs, and ethnic identification. The central estimand is the *conditional demographic trajectory* generated by a specified set of assumptions—not a prediction of future population composition. Phase 1 establishes the repository architecture, official data ingestion pipelines, ethnic category harmonisation for England and Wales (Census 2021), and a minimal tested projection engine. All illustrative outputs in this phase use clearly labelled placeholder parameters and must not be interpreted as empirical projections. This is a *methods and software design* contribution; empirical calibration and historical validation are required before substantive long-run scenario analysis.

Keywords: population projection, ethnicity, migration, cohort-component model, United Kingdom, demographic scenarios

1 Introduction

Population projections by ethnic group are analytically important and methodologically demanding. Ethnicity in the United Kingdom is measured through self-identification on

the census; categories have changed across census rounds; and the demographic behaviour associated with ethnic groups—fertility, mortality, migration, partnership, and identity transmission—differs in ways that standard national projections do not capture (Haskey, 2002; Rees et al., 2011, 2017).

Substantial prior work demonstrates both the feasibility and the sensitivity of long-run ethnic composition trajectories to assumptions about fertility convergence and migration, including Wohland et al. (2010), Coleman (2010), and the NEWETHPOP system (NEWETHPOP, 2013). This project builds on that tradition with an open-source pipeline updated for Census 2021/2022 base populations, explicit preregistration of scenarios, separate immigration and emigration modules, and planned ethnic identity transition matrices estimated from census longitudinal change.

The research question is:

How does the projected ethnic composition of the United Kingdom change between 2022 and 2122 under alternative migration and demographic scenarios?

The answer is always conditional: under assumptions A , the model produces trajectory T_A . The system never claims that “the UK will have” a particular ethnic composition.

1.1 Scope and limitations (Phase 1)

Phase 1 delivers the system architecture, data pipelines, harmonisation schema, and a minimal tested projection engine. **No parameters are calibrated to UK data in this phase.** The illustrative scenario uses placeholder values and is watermarked:

Methodological demonstration only. Not an empirical projection.

What Phase 1 can establish: the model structure is auditable and reproducible; Census 2021 ethnicity data for England and Wales are retrievable via official APIs; and the harmonisation framework is specified. What it cannot yet establish: any empirical statement about future UK ethnic composition, uncertainty quantification, or historical forecast accuracy.

1.2 Phase 1 implementation status

Table 1 summarises which model components are implemented versus planned.

Table 1: Phase 1 implementation status

Component	Status	Notes
Cohort ageing and survival	Implemented	Placeholder life tables
Immigration / emigration (fixed flows)	Implemented	Not yet rate-based
Ethnic identity matrix Q	Implemented	Diagonal (fixed identity)
Births from maternal ASFR	Implemented	Sex ratio at birth applied
Child generation from parent nativity	Implemented	Simplified rules
Partnership-weighted fertility	Partial	Weights only; males not required in stock
Fertility convergence	Planned	Parameters stored, not applied
Generation transition at ageing	Planned	Phase 3
Probabilistic uncertainty	Planned	Phase 5
Historical hindcast validation	Planned	Phase 4

2 Conceptual framework

2.1 Ethnicity as self-identified and changeable

Ethnicity is treated as a self-identified social category, distinct from nationality, citizenship, religion, ancestry, and country of birth (ONS, 2021, 2023). A person may report a different ethnic identity at different censuses. Country of birth is never used as a direct proxy for ethnicity; where parental country of birth informs generation classification, the mapping is empirically estimated and documented.

2.2 Population state space

The population at year t is represented as:

$$N_{n,e,g,s,a,t} \tag{1}$$

where n indexes UK nation, e ethnic group, g migrant generation, s sex, and a age (0–100+). Generation categories are:

1. foreign-born, arrived as adult;
2. foreign-born, arrived as child;
3. UK-born with two foreign-born parents;
4. UK-born with one foreign-born parent;
5. UK-born with two UK-born parents.

Generation is not inferred from ethnicity.

2.3 Geographic scope

England, Wales, Scotland, and Northern Ireland are modelled separately. Internal migration flows sum to zero at UK level. Census base dates differ: England, Wales, and Northern Ireland use Census 2021 (21 March 2021); Scotland uses Census 2022 (20 March 2022). A harmonised mid-2022 base will adjust census-date counts using official mid-year population estimates (Phase 2).

3 Projection model

3.1 Cohort-component transition

For existing cohorts, the core transition equation is:

$$N_{n,e,g,s,a+1,t+1} = \sum_{e',g'} N_{n,e',g',s,a,t} \cdot S_{n,e',g',s,a,t} \cdot Q_{e' \rightarrow e,t} + I_{n,e,g,s,a,t} - O_{n,e,g,s,a,t} + M_{n,e,g,s,a,t}^{int} \quad (2)$$

where S is survival probability, Q is the ethnic-identification transition matrix (operating on e only in Phase 1), I is immigration, O is emigration, and M^{int} is net internal movement between UK nations. Deaths D are implicit in $(1 - S)$ and appear explicitly in the accounting identity (Section 6).

3.2 Birth model

Births depend on maternal age-specific fertility rates (ASFR), not aggregate TFR alone:

$$B_{n,e,g,s,t} = \sum_{a,e_m,g_m,e_p,g_p} N_{n,e_m,g_m,a,t}^F \cdot f_{n,e_m,g_m,a,t} \cdot P(e_p, g_p \mid e_m, g_m, n, t) \cdot P(e, g \mid e_m, e_p, g_m, g_p, n, t) \cdot P(s \mid t) \quad (3)$$

where $P(s \mid t)$ is the sex ratio at birth. Children are not automatically assigned the mother's ethnicity; $P(e, g \mid \cdot)$ is estimated from census parental-ethnicity tables (Phase 7) or specified as a scenario assumption. In Phase 1, placeholder child-identification probabilities are used and labelled accordingly.

Partnership affects fertility inputs only: the model does not include a partnership state dimension. Two-sex couple formation is deferred to Phase 7.

3.3 Fertility convergence (planned)

Group-specific fertility will converge toward the national schedule:

$$f_{e,g,a,t} = f_{a,t}^{UK} + \lambda_{e,g,t} (f_{e,g,a}^{initial} - f_{a,t}^{UK}), \quad \lambda_{e,g,t} = \exp(-\kappa_{e,g} d) \quad (4)$$

where d is duration of residence in the UK. Six specifications are planned; Phase 1 applies static ASFR only.

3.4 Migration

Immigration and emigration are modelled separately at nation level:

$$M_{n,e,g,s,a,t}^{net} = I_{n,e,g,s,a,t} - O_{n,e,g,s,a,t} \quad (5)$$

Migration scenarios specify measurable parameters: volumes, age and sex distributions, ethnic composition, country of birth, entry route, and subsequent emigration probability. Historical calibration will use admin-based migration estimates (ABME) from 2020 onward, with IPS/LTIM used for earlier periods subject to documented comparability breaks (ONS, 2024a).

4 Data sources and harmonisation

4.1 Official data retrieved (Phase 1)

Table 2 summarises data sources. Phase 1 successfully retrieved Census 2021 ethnicity $\times$ age $\times$ sex for England and Wales via the ONS Beta API (RM032) and Nomis (dataset NM_2132, table C2021RM032). Both sources provide age in **broad bands** (six categories), not single year of age; disaggregation to a full age structure requires statistical methods using mid-year population estimates (Phase 2).

Table 2: Primary official data sources and retrieval status

Parameter	Source	Phase 1 status	Phase
Ethnicity $\times$ age (E&W)	ONS RM032 / Nomis NM_2132	Retrieved	1
Ethnicity $\times$ age (Scotland)	NRS Census 2022	Not retrieved	6
Ethnicity $\times$ age (NI)	NISRA DT-0035	Not retrieved	6
Mid-year estimates 2022	ONS MYE	Pending	2
International migration	ONS ABME / LTIM	Pending	4
National life tables	ONS	Pending	3
Census 2001/2011 ethnicity	Nomis	Pending	4

Nomis internal API codes (1–19) for ethnic categories differ from Census output codes (10001–50002); the harmonisation table records both identifiers.

4.2 Harmonisation

Two parallel ethnic classifications are maintained: a five-group broad classification (**white**; **mixed**; **asian**; **black**; **other**) and a detailed classification using harmonisable Census 2021 England/Wales categories. A version-controlled mapping table records source country, census year, original category, API codes, harmonised category, mapping rationale, comparability quality, and uncertainty flag. Scotland, Northern Ireland, and historical census years are pending (Phases 4–6).

5 Scenario design and uncertainty

Scenarios are defined in YAML configuration files specifying measurable parameters. Policy scenarios modify migration volumes, route shares, and composition—not vague labels such as “strict” or “liberal.”

Two modes are planned:

- **Deterministic:** explicit assumptions produce exact conditional trajectories.
- **Probabilistic (Phase 5):** parameters drawn from $\theta^{(r)} \sim p(\theta)$ with $\geq 2,000$ Monte Carlo draws (ONS, 2024b).

Deterministic scenario ranges are never called confidence intervals.

6 Validation strategy

6.1 Accounting validation

Every projection year verifies:

$$P_{t+1} = P_t + B_t - D_t + I_t - O_t + M_t^{int} \quad (6)$$

where D_t denotes deaths and M_t^{int} denotes net internal migration (distinct symbols throughout). Additional checks: non-negative counts, valid probabilities, ethnic shares summing to one, and internal migration summing to zero at UK level.

6.2 Historical validation

Hindcasts from Census 2001 to 2011 and 2011 to 2021/2022 will be conducted before any 75- or 100-year outputs are reported. Metrics include MAE, RMSE, share error, and

coverage. MAPE is reported only for groups above a minimum population threshold.

7 Phase 1 results: system demonstration

The Phase 1 illustrative scenario uses a synthetic population of 1 million distributed across England and Wales with approximate Census 2021 broad ethnic shares. All demographic parameters are placeholders. Projections run to 2050 in the stored output (the research design specifies 2122, but long-horizon outputs require validated parameters).

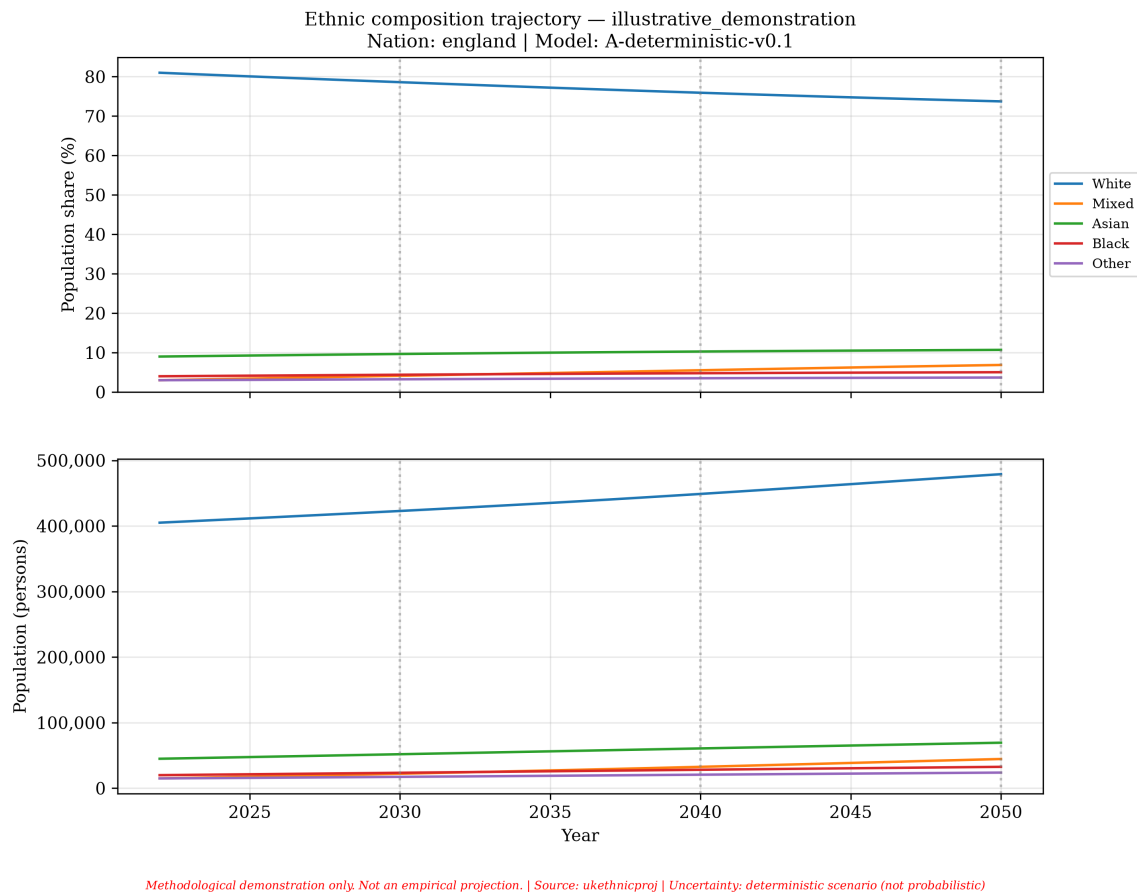


Figure 1: Ethnic population shares under the illustrative demonstration scenario (England only; Wales is included in the scenario but excluded from this figure). Projection horizon: 2022–2050. **Methodological demonstration only. Not an empirical projection.** Parameters are placeholders, not calibrated to UK data.

Figure 1 shows conditional trajectories under placeholder assumptions only. No inference about the actual future ethnic composition of the UK should be drawn from this figure.

8 What this study can and cannot say

Can say (Phase 1):

- The projection system architecture is specified, implemented, and tested.
- Census 2021 ethnicity data for England and Wales are retrievable with documented provenance.
- Under clearly labelled placeholder assumptions, the model satisfies the population accounting identity.

Cannot yet say:

- How the UK’s ethnic composition will change by 2122.
- Which assumptions contribute most to uncertainty.
- Whether the model would have accurately projected Census 2011 or 2021.
- Nation-specific projections for Scotland and Northern Ireland.

9 Conclusion

This paper presents the Phase 1 architecture of a population-projection system for conditional ethnic demographic trajectories under alternative scenarios. The system treats ethnicity as self-identified and changeable, models immigration and emigration separately, and requires historical validation before long-run outputs. Empirical calibration, probabilistic uncertainty, and full UK coverage follow in subsequent phases.

Data and code availability

Source code, scenario configurations, harmonisation tables, and data manifests are available in the project repository. Raw API responses are cached with SHA-256 checksums under `data/raw/`. Census microdata are not used; all inputs are aggregate official tables under the Open Government Licence v3.0.

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This is unfunded independent research. The author declares no conflicts of interest.

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